ASSIGNMENT X - BACK END / GROUP PROJECT

32-bit RISC-V processor

Floorplanning through routing, timing analysis and physical checks

Team member	Registration number	Lab slot
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Five elastic stages: instruction fetch, decode, execute, memory and writeback. The physical design includes the processor and its iterative multiply/divide unit. Instruction/data memory, caches, the accelerator and chip IO pads are outside this GDS boundary.

Implementation platform: public Nangate45 / FreePDK45, typical Liberty corner at 1.1 V and 25 C. Clock target: 10 ns. Results are measured from saved tool runs; limitations are identified alongside the corresponding checks.

Prepared against the supplied assignment question paper. Screenshots are actual tool windows captured on an isolated Linux X11 display; terminal figures explicitly review recorded logs. Full source, logs and databases accompany the project.

Implementation outcome and boundaries

Metric	Result
Physical top	rv32_pipeline including rv32_muldiv
Die dimensions	292.425 x 292.425 um
Core dimensions	272.270 x 270.200 um; area 73567.354 um2
Final cell area / utilization	36606 um2 / 50% (tool-rounded; excludes filler inflation)
Post-route setup / hold slack	6.31 ns / 0.02 ns; extracted RC, typical corner
Detailed-router design-rule markers	0
Maximum-capacitance violations	0 after final routed-RC extraction
Independent GDS DRC	0 markers; see physical-verification section
LVS status	PASS: routed processor topology and all 443 named top ports match

Tools and reproducible environment

Tool / input	Version or identity
yosys	Yosys 0.9 (git sha1 1979e0b)
iverilog	Icarus Verilog version 11.0 (stable) ()
verilator	Verilator 4.038 2020-07-11 rev v4.036-114-g0cd4a57ad
openroad	v2.0-17598-ga008522d8
klayout	KLayout 0.30.12
gcc	riscv64-unknown-elf-gcc () 10.2.0

Platform checkout: 6101364b2d7909dd797e1e3e7f80695401cfa4e4. The official Ubuntu 22.04 OpenROAD release and its SHA256 are recorded in reports/toolchain_installation.json. No commercial libraries were required.

Toolchain verification

Actual version-command output captured in xterm. The installation package and platform revision are pinned in the project.

```
+ cat build/evidence/tools_review.txt
OPEN-SOURCE TOOLCHAIN - actual installed versions

yosys: Yosys 0.9 (git sha1 1979e0b)
iverilog: Icarus Verilog version 11.0 (stable) ()
verilator: Verilator 4.038 2020-07-11 rev v4.036-114-g0cd4a57ad
openroad: v2.0-17598-ga008522d8
klayout: KLayout 0.30.12
gcc: riscv64-unknown-elf-gcc () 10.2.0
platform_commit: 6101364b2d7909dd797e1e3e7f80695401cfa4e4

Platform: public Nangate45 / FreePDK45
Physical scope: core only; no SoC SRAM or pad ring
Analysis: typical corner; not foundry signoff
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Floorplanning

A square floorplan targets 40% initial cell utilization with approximately 10 um core margins. Rows snap to the 0.19 x 1.4 um standard-cell site. Pins are distributed on metal5/metal6. The extra whitespace accommodates repair buffers and routing; no SRAM macros or IO pads are placed.

Metric	Measured result
Cell area	29686.0 um2 (rounded by tool)
Cell utilization	40% (rounded by tool)
Setup slack / hold slack	-5.64 ns / 0.03 ns
Database / reports	build/asic/2_floorplan.odb, .def, .sdc, _setup.rpt and _hold.rpt

Representative stage command: initialize_floorplan -utilization 40 -aspect_ratio 1 -core_space 10

Floorplanning - tool results

Recorded stage metrics and the beginning of a real setup path from the corresponding STA report. Refer to the full report for every path arc.

```
+ cat build/evidence/2_floorplan_review.txt
PHYSICAL STAGE: 2_floorplan
Command: openroad -exit asic/open/flow.tcl
Source: build/asic/flow.log
Die: [0.0, 0.0, 292.425, 292.425] um
Core: [10.07, 11.2, 282.34, 281.4] um

cell_area_um2_rounded: 29686.0
utilization_percent_rounded: 40
setup_wns_ns: -5.64
hold_wns_ns: 0.03

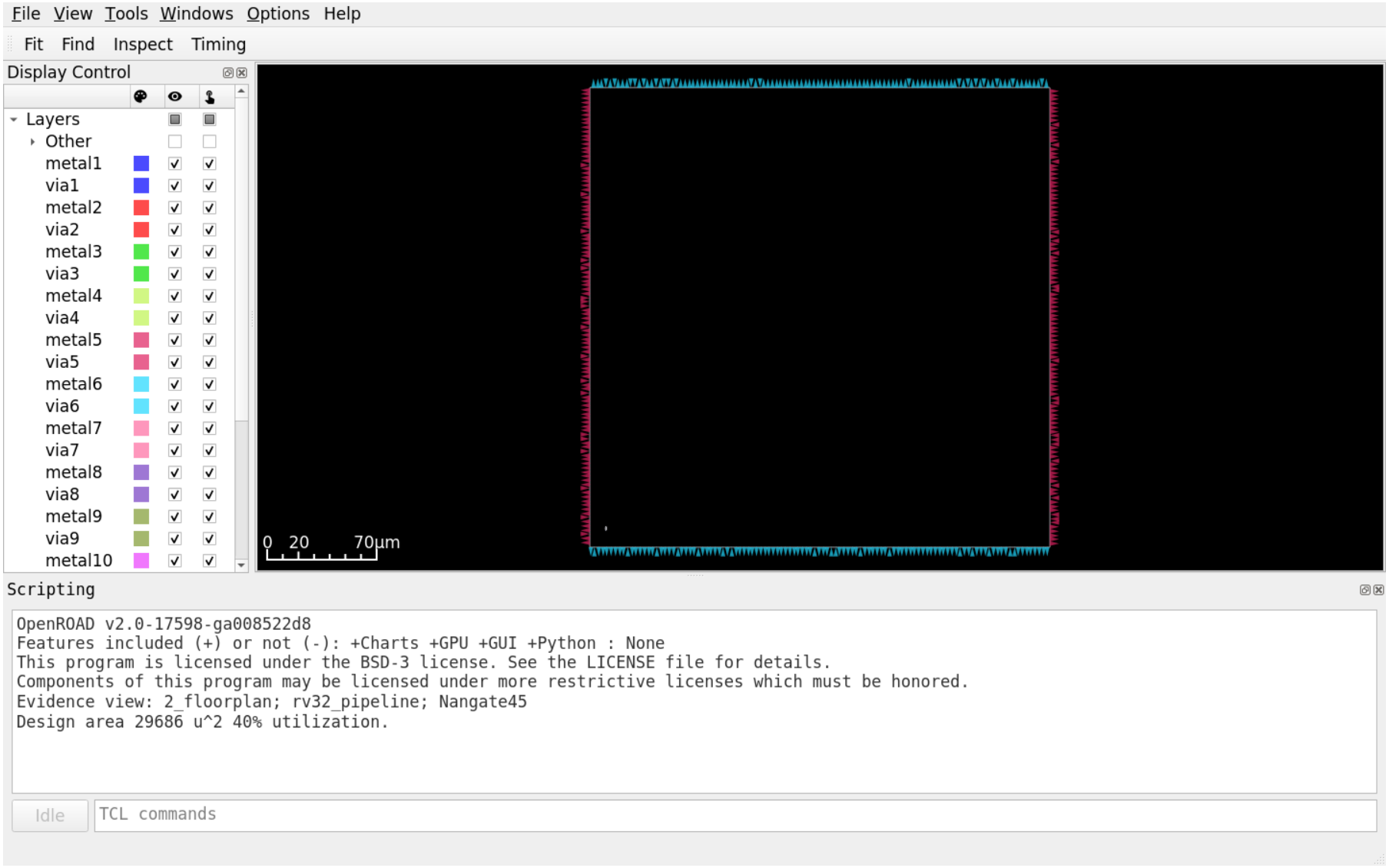
Startpoint: rst (input port clocked by core_clk)
Endpoint: _34217_ (rising edge-triggered flip-flop clocked by core_clk)
Path Group: core_clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   1.00    1.00 v input external delay
   0.00    1.00 v rst (in)
   4.30    5.30 ^ _16598_/ZN (INV_X1)
   8.09   13.39 v _17301_/ZN (NAND2_X1)
   1.85   15.24 ^ _29018_/ZN (NOR2_X1)
   0.10   15.33 v _29996_/ZN (OAI211_X1)
   0.07   15.40 ^ _29997_/ZN (OAI211_X1)
   0.00   15.40 ^ _34217_/D (DFF_X1)
          15.40   data arrival time
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Floorplanning - physical view

Actual OpenROAD view of 2_floorplan.odb. The stage database and constraints are loaded from the implementation output; no layout geometry is drawn for this report.



Power planning

Global VDD/VSS connections feed metal1 follow-pin rails. Metal4 stripes are 0.48 um wide at 40 um pitch; metal7 straps are 1.4 um wide at 30 um pitch. A metal6/metal7 ring uses 1.4 um widths, 1.0 um spacing and 3.0 um core offset. Via connections join metal1-metal4, metal4-metal7 and metal6-metal7. The final power-grid check verifies connectivity, not package supply or IR-drop signoff.

Metric	Measured result
Cell area	29686.0 um2 (rounded by tool)
Cell utilization	40% (rounded by tool)
Setup slack / hold slack	-5.64 ns / 0.03 ns
Database / reports	build/asic/3_power.odb, .def, .sdc, _setup.rpt and _hold.rpt

Representative stage command: pdngen -failed_via_report build/asic/pdn_failed_vias.rpt

Power planning - tool results

Recorded stage metrics and the beginning of a real setup path from the corresponding STA report. Refer to the full report for every path arc.

```
+ cat build/evidence/3_power_review.txt
PHYSICAL STAGE: 3_power
Command: openroad -exit asic/open/flow.tcl
Source: build/asic/flow.log
Die: [0.0, 0.0, 292.425, 292.425] um
Core: [10.07, 11.2, 282.34, 281.4] um

cell_area_um2_rounded: 29686.0
utilization_percent_rounded: 40
setup_wns_ns: -5.64
hold_wns_ns: 0.03

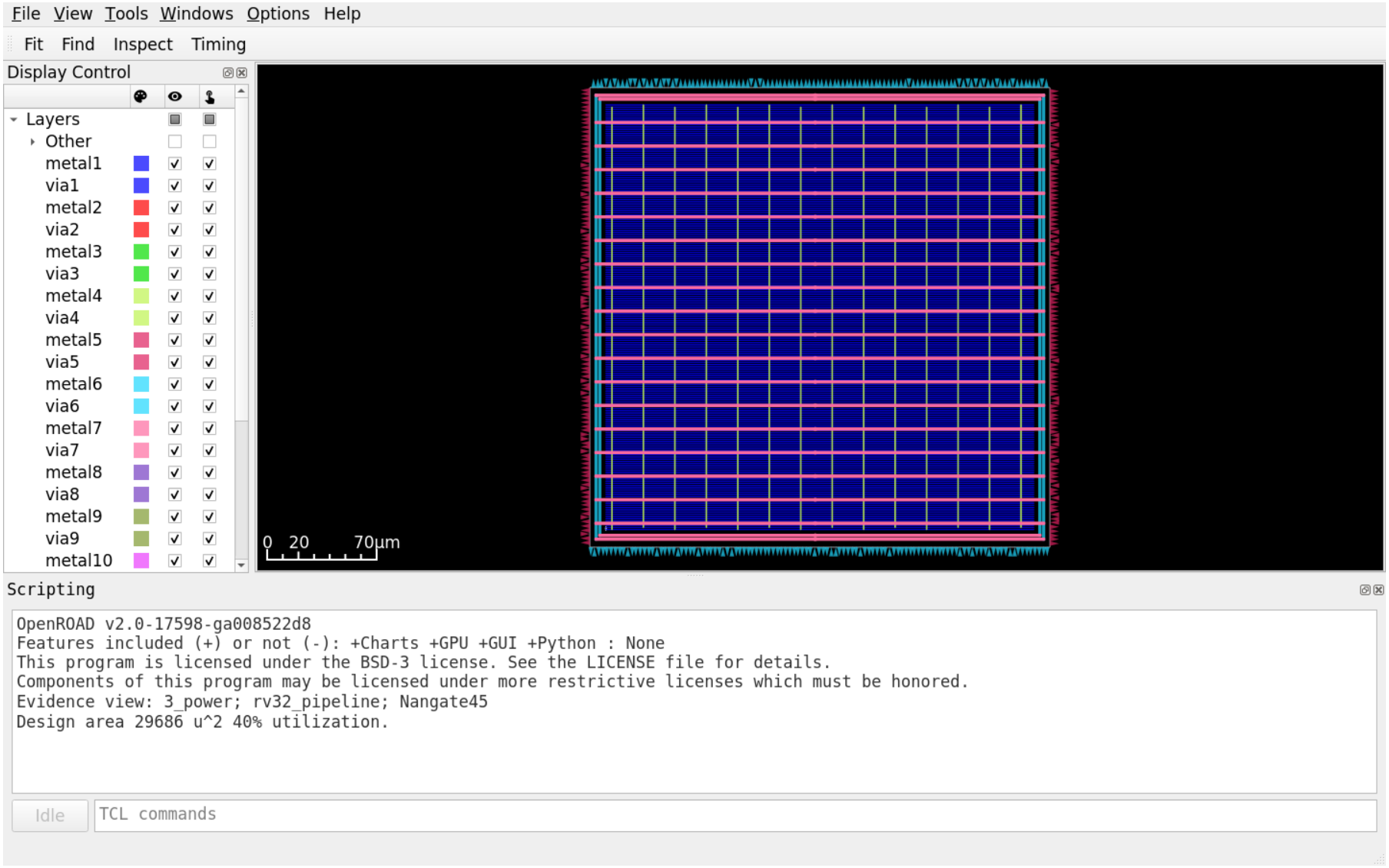
Startpoint: rst (input port clocked by core_clk)
Endpoint: _34217_ (rising edge-triggered flip-flop clocked by core_clk)
Path Group: core_clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   1.00    1.00 v input external delay
   0.00    1.00 v rst (in)
   4.30    5.30 ^ _16598_/ZN (INV_X1)
   8.09   13.39 v _17301_/ZN (NAND2_X1)
   1.85   15.24 ^ _29018_/ZN (NOR2_X1)
   0.10   15.33 v _29996_/ZN (OAI211_X1)
   0.07   15.40 ^ _29997_/ZN (OAI211_X1)
   0.00   15.40 ^ _34217_/D (DFF_X1)
          15.40   data arrival time
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Power planning - physical view

Actual OpenROAD view of 3_power.odb. The stage database and constraints are loaded from the implementation output; no layout geometry is drawn for this report.



Placement

Routability-driven global placement is followed by electrical repair, resizing and legalization. One-site padding reserves space between cells. Initial synthesis slew/capacitance violations motivate repair. The resulting area increase is reported separately from the synthesis area; placement is checked for overlaps and row/site legality.

Metric	Measured result
Cell area	35694.0 um2 (rounded by tool)
Cell utilization	49% (rounded by tool)
Setup slack / hold slack	6.26 ns / 0.03 ns
Database / reports	build/asic/4_placement.odb, .def, .sdc, _setup.rpt and _hold.rpt

Representative stage command: global_placement -density 0.55 -routability_driven; detailed_placement

Placement - tool results

Recorded stage metrics and the beginning of a real setup path from the corresponding STA report. Refer to the full report for every path arc.

```
+ cat build/evidence/4_placement_review.txt
PHYSICAL STAGE: 4_placement
Command: openroad -exit asic/open/flow.tcl
Source: build/asic/flow.log
Die: [0.0, 0.0, 292.425, 292.425] um
Core: [10.07, 11.2, 282.34, 281.4] um

cell_area_um2_rounded: 35694.0
utilization_percent_rounded: 49
setup_wns_ns: 6.26
hold_wns_ns: 0.03

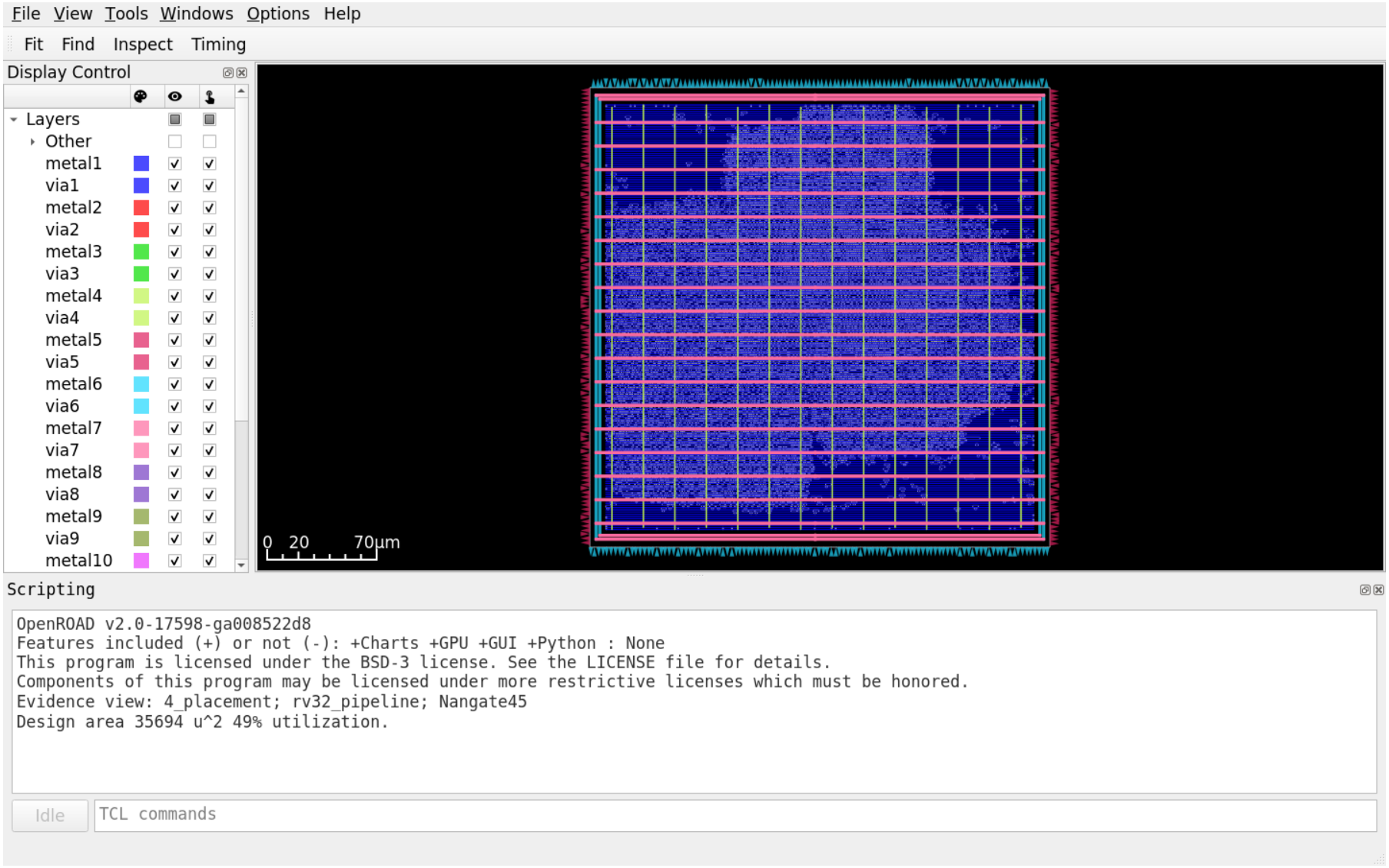
Startpoint: dp (input port clocked by core_clk)
Endpoint: _33164_ (rising edge-triggered flip-flop clocked by core_clk)
Path Group: core_clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   1.00    1.00 v input external delay
   0.00    1.00 v dp (in)
   0.10    1.10 v input34/Z (BUF_X1)
   0.06    1.15 ^ _17290_/ZN (OAI211_X1)
   0.06    1.21 ^ _17291_/ZN (AND2_X1)
   0.03    1.25 v _17773_/ZN (NAND3_X2)
   0.03    1.28 ^ _17774_/ZN (INV_X1)
   0.02    1.30 v _17793_/ZN (OAI21_X4)
   0.12    1.42 v _17840_/ZN (OR4_X4)
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Placement - physical view

Actual OpenROAD view of 4_placement.odb. The stage database and constraints are loaded from the implementation output; no layout geometry is drawn for this report.



Clock tree synthesis

TritonCTS builds a clustered tree from the clk input. Clock inverters and clock nets are repaired, the new buffers are legalized and clocks become propagated for timing. The actual Clock Tree Viewer and stage log retain the clock-buffer hierarchy and counts. The original 0.2 ns hold uncertainty exhausted repair limits; the revised hold budget is 0.05 ns while setup stays 0.2 ns. This is a disclosed constraint revision, not closure against the earlier hold budget.

Metric	Measured result
Cell area	36574.0 um2 (rounded by tool)
Cell utilization	50% (rounded by tool)
Setup slack / hold slack	6.47 ns / 0.02 ns
Database / reports	build/asic/5_cts.odb, .def, .sdc, _setup.rpt and _hold.rpt

Representative stage command: clock_tree_synthesis -root_buf CLKBUF_X3 -buf_list {CLKBUF_X1 CLKBUF_X2 CLKBUF_X3}

Clock tree synthesis - tool results

Recorded stage metrics and the beginning of a real setup path from the corresponding STA report. Refer to the full report for every path arc.

```
Core: [10.07, 11.2, 282.34, 281.4] um

cell_area_um2_rounded: 36574.0
utilization_percent_rounded: 50
setup_wns_ns: 6.47
hold_wns_ns: 0.02

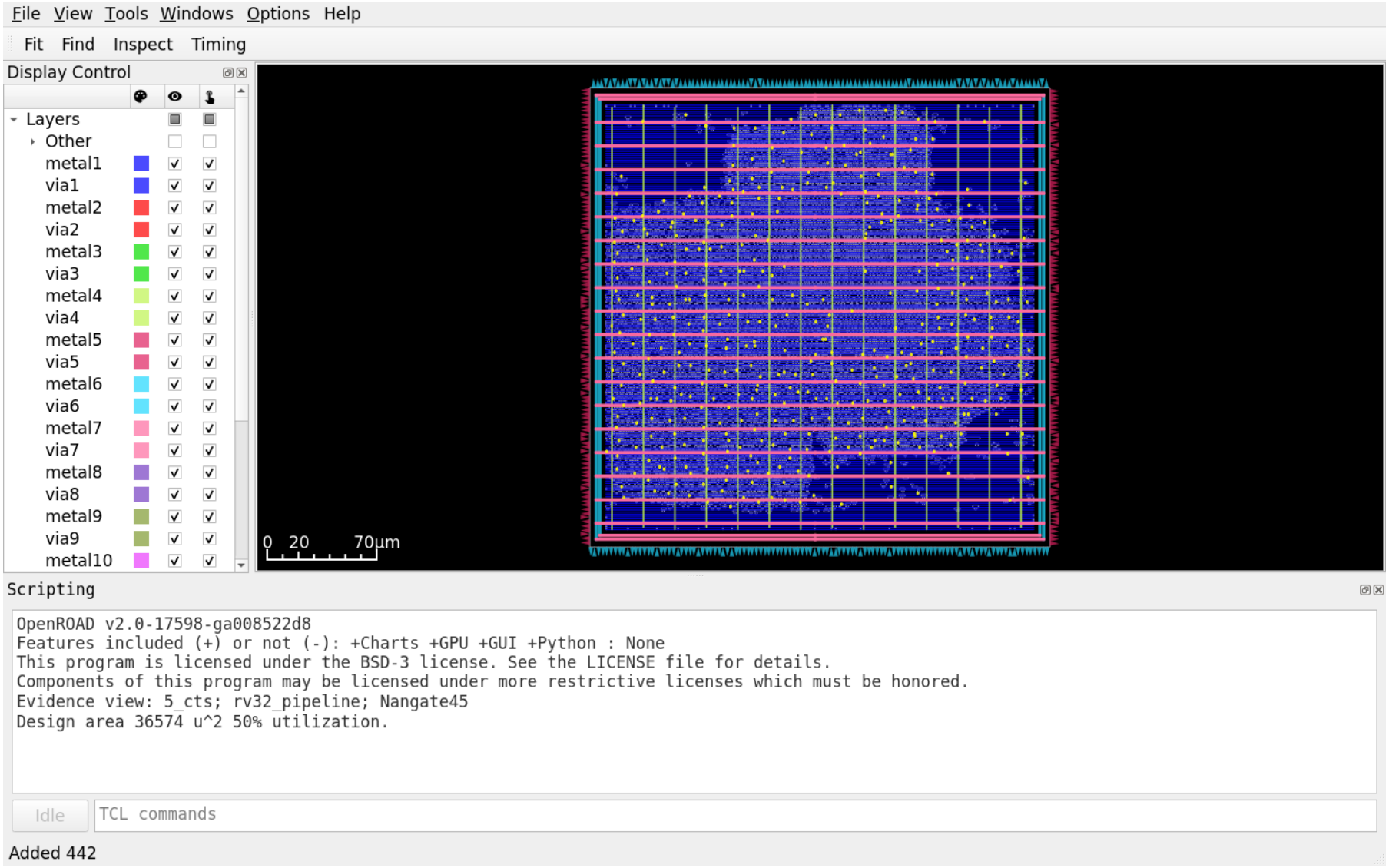
Startpoint: dp (input port clocked by core_clk)
Endpoint: _33164_ (rising edge-triggered flip-flop clocked by core_clk)
Path Group: core_clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clk (rise edge)
   0.00    0.00   clock network delay (propagated)
   1.00    1.00 v input external delay
   0.00    1.00 v dp (in)
   0.10    1.10 v input34/Z (BUF_X1)
   0.06    1.15 ^ _17290_/ZN (OAI211_X1)
   0.06    1.21 ^ _17291_/ZN (AND2_X1)
   0.03    1.24 v _17773_/ZN (NAND3_X2)
   0.03    1.28 ^ _17774_/ZN (INV_X1)
   0.02    1.30 v _17793_/ZN (OAI21_X4)
   0.12    1.42 v _17840_/ZN (OR4_X4)
[INFO CTS-0018]     Created 442 clock buffers.
[INFO CTS-0015]     Created 442 clock nets.
[INFO CTS-0100]  Leaf buffers 405
[INFO CTS-0102]  Path depth 3 - 4
[INFO RSZ-0033] No hold violations found.
[INFO RSZ-0033] No hold violations found.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Clock tree synthesis - physical view

Actual OpenROAD view of 5_cts.odb. The stage database and constraints are loaded from the implementation output; no layout geometry is drawn for this report.



Global routing

Signals use metal2-metal6 and clocks metal4-metal6 to limit exposure to thick upper-metal antenna geometry. Routing capacities are reduced by 30% to reserve resources. The logged global-congestion table records the layer-by-layer overflow and resource use; the initial global-route guide wirelength is recorded in asic_metrics.json. Global routing writes guides for detailed routing. Timing here uses global-route parasitic estimates; it is not the extracted final timing.

Metric	Measured result
Cell area	36606.0 um2 (rounded by tool)
Cell utilization	50% (rounded by tool)
Setup slack / hold slack	6.41 ns / 0.02 ns
Database / reports	build/asic/6_global_route.odb, .def, .sdc, _setup.rpt and _hold.rpt

Representative stage command: global_route -congestion_iterations 100 -verbose

Global routing - tool results

Recorded stage metrics and the beginning of a real setup path from the corresponding STA report. Refer to the full report for every path arc.

```
+ cat build/evidence/6_global_route_review.txt
PHYSICAL STAGE: 6_global_route
Command: openroad -exit asic/open/flow.tcl
Source: build/asic/flow.log
Die: [0.0, 0.0, 292.425, 292.425] um
Core: [10.07, 11.2, 282.34, 281.4] um

cell_area_um2_rounded: 36606.0
utilization_percent_rounded: 50
setup_wns_ns: 6.41
hold_wns_ns: 0.02

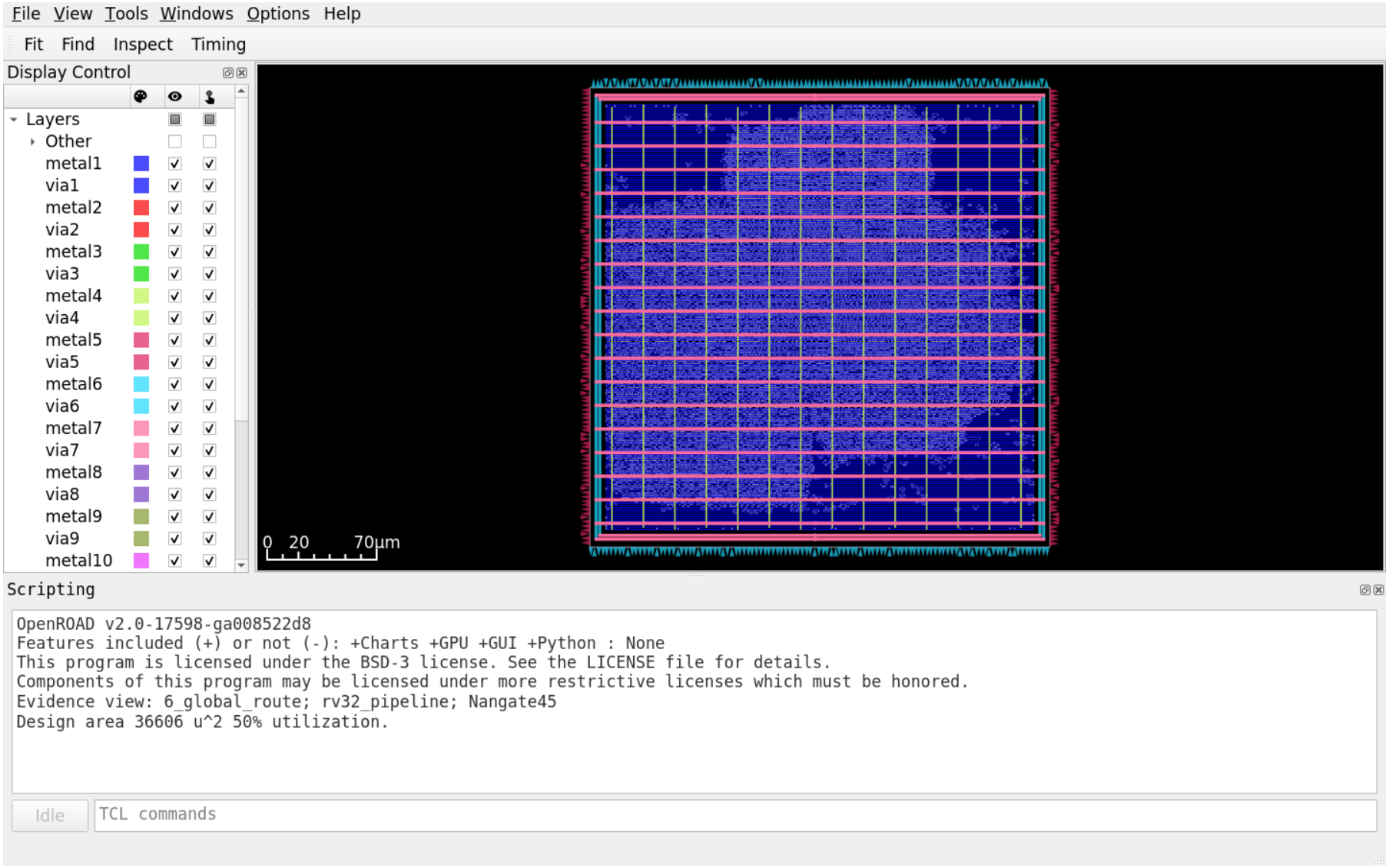
Startpoint: dp (input port clocked by core_clk)
Endpoint: _33164_ (rising edge-triggered flip-flop clocked by core_clk)
Path Group: core_clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clk (rise edge)
   0.00    0.00   clock network delay (propagated)
   1.00    1.00 v input external delay
   0.00    1.00 v dp (in)
   0.09    1.09 v input34/Z (BUF_X1)
   0.06    1.15 ^ _17290_/ZN (OAI211_X1)
   0.06    1.21 ^ _17291_/ZN (AND2_X1)
   0.03    1.24 v _17773_/ZN (NAND3_X2)
   0.03    1.27 ^ _17774_/ZN (INV_X1)
   0.03    1.30 v _17793_/ZN (OAI21_X4)
   0.12    1.42 v _17840_/ZN (OR4_X4)
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Global routing - physical view

Actual OpenROAD view of 6_global_route.odb. The stage database and constraints are loaded from the implementation output; no layout geometry is drawn for this report.



Detailed routing and extraction

Detailed routing resolves track assignment and vias using the guides. Standard-cell filler masters complete row gaps. The flow exports routed DEF and mapped Verilog, extracts routed RC with OpenRCX, writes/reads SPEF and reports setup/hold with propagated clocks. The final cell area is a logic/repair-area metric, not the die area or a full-chip pad/memory area.

Metric	Measured result
Cell area	36606.0 um2 (rounded by tool)
Cell utilization	50% (rounded by tool)
Setup slack / hold slack	6.31 ns / 0.02 ns
Database / reports	build/asic/7_final.odb, .def, .sdc, _setup.rpt and _hold.rpt

Representative stage command: detailed_route; extract_parasitics; write_spef; read_spef

Detailed routing and extraction - tool results

Recorded stage metrics and the beginning of a real setup path from the corresponding STA report. Refer to the full report for every path arc.

```
PHYSICAL STAGE: 7_final
Command: openroad -exit asic/open/flow.tcl
Source: build/asic/flow.log
Die: [0.0, 0.0, 292.425, 292.425] um
Core: [10.07, 11.2, 282.34, 281.4] um

cell_area_um2_rounded: 36606.0
utilization_percent_rounded: 50
setup_wns_ns: 6.31
hold_wns_ns: 0.02

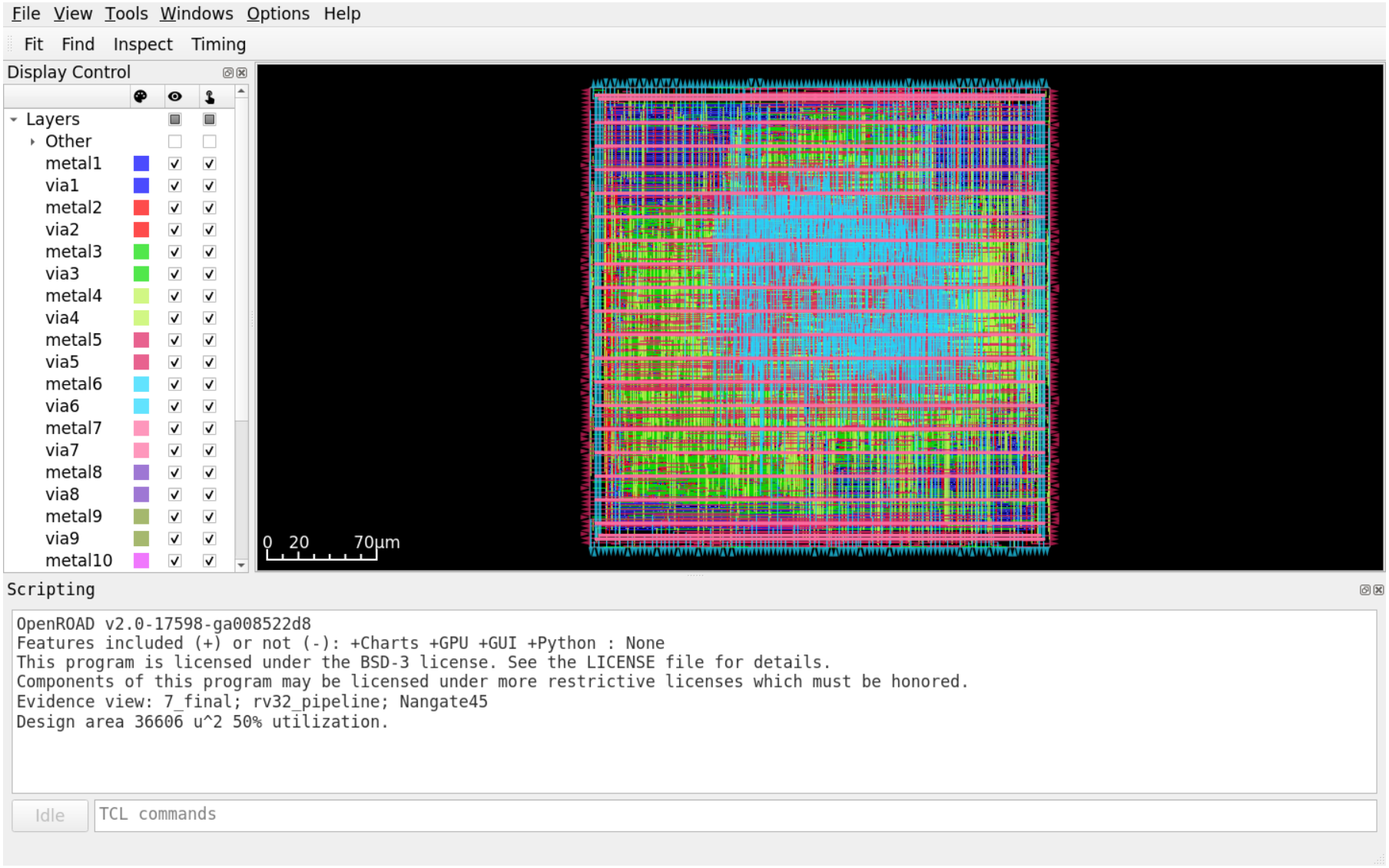
Startpoint: dp (input port clocked by core_clk)
Endpoint: _33164_ (rising edge-triggered flip-flop clocked by core_clk)
Path Group: core_clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clk (rise edge)
   0.00    0.00   clock network delay (propagated)
   1.00    1.00 v input external delay
   0.00    1.00 v dp (in)
   0.09    1.09 v input34/Z (BUF_X1)
   0.06    1.15 ^ _17290_/ZN (OAI211_X1)
   0.06    1.21 ^ _17291_/ZN (AND2_X1)
   0.03    1.25 v _17773_/ZN (NAND3_X2)
   0.03    1.28 ^ _17774_/ZN (INV_X1)
   0.03    1.30 v _17793_/ZN (OAI21_X4)
   0.12    1.42 v _17840_/ZN (OR4_X4)
Router DRVs: 0
FLOW_COMPLETE: True
+ set +x

Recorded log review complete. Original logs retained in build/.
```

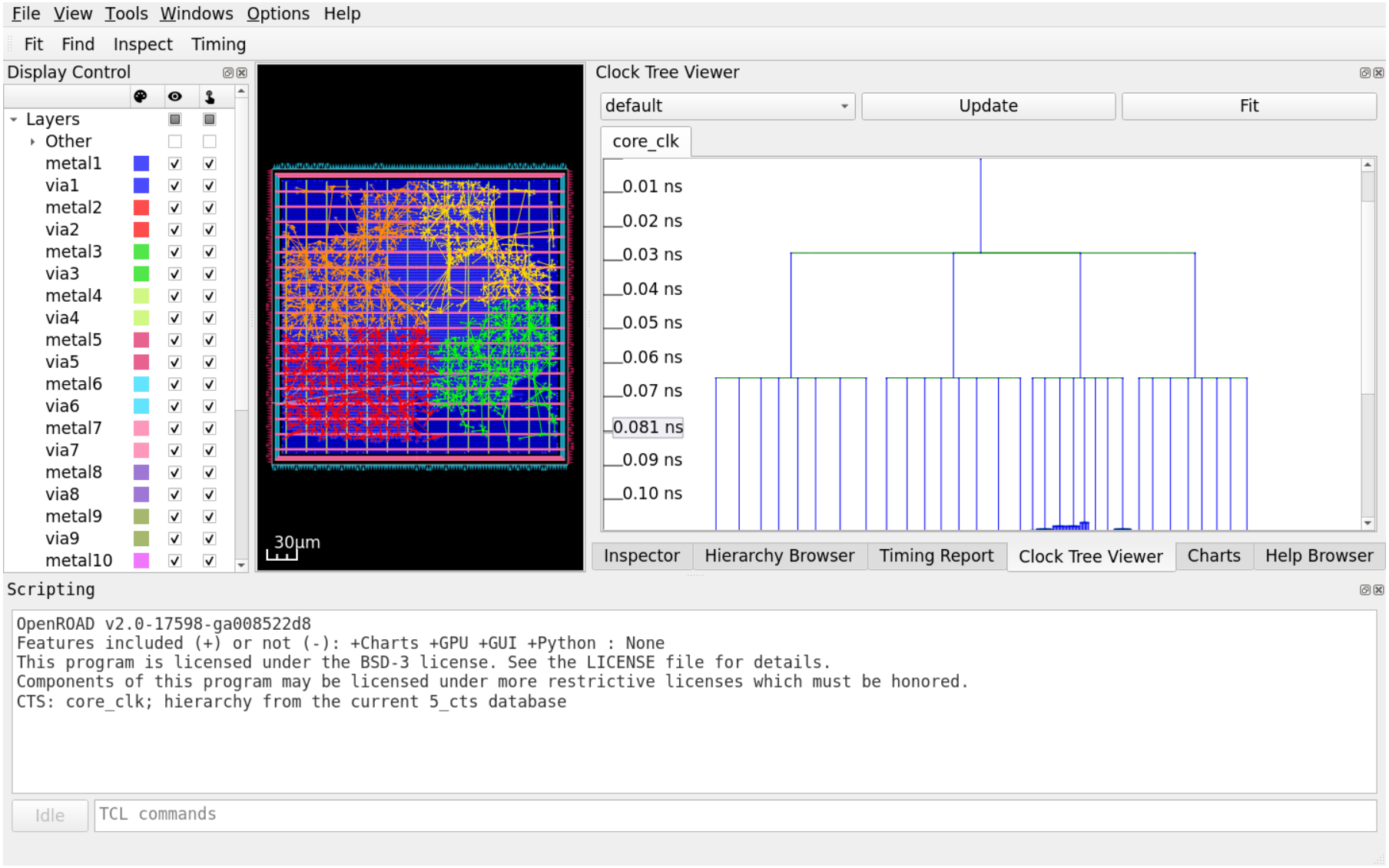
Detailed routing and extraction - physical view

Actual OpenROAD view of 7_final.odb. The stage database and constraints are loaded from the implementation output; no layout geometry is drawn for this report.



Clock tree hierarchy in OpenROAD

Actual Clock Tree Viewer for core_clk after CTS. The buffer hierarchy supplements the placed clock-buffer view and recorded tree metrics.



Hold-budget revision and failed attempts

Both failed logs are preserved. The final run uses a disclosed 50 ps hold uncertainty; it does not establish closure against the original 200 ps hold budget.

```
+ cat build/evidence/cts_constraint_revision_review.txt
CTS CONSTRAINT REVISION - preserved failed attempts
Original uncertainty: setup=0.2 ns and hold=0.2 ns
Final uncertainty: setup=0.2 ns and hold=0.05 ns

Source: build/evidence/cts_first_attempt.log
delta HPWL                     3 %

[INFO RSZ-0098] No setup violations found
[INFO RSZ-0046] Found 2663 endpoints with hold violations.
[WARNING RSZ-0066] Unable to repair all hold violations.
[INFO RSZ-0032] Inserted 3774 hold buffers.
[ERROR RSZ-0060] Max buffer count reached.
Error: flow.tcl, 70 RSZ-0060

Source: build/evidence/cts_second_attempt.log
_21957_/Z                             120.85  123.09   -2.24 (VIOLATED)

[INFO RSZ-0098] No setup violations found
[INFO RSZ-0046] Found 2663 endpoints with hold violations.
[WARNING RSZ-0066] Unable to repair all hold violations.
[INFO RSZ-0032] Inserted 9432 hold buffers.
[ERROR RSZ-0060] Max buffer count reached.
Error: flow.tcl, 71 RSZ-0060

Final run: no hold violations before repair under the revised budget.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Timing analysis and closure assessment

Stage	Cell area (um2)	Util.	Setup slack (ns)	Hold slack (ns)
1_synth	29686.0	N/A	-5.64	0.03
4_placement	35694.0	49%	6.26	0.03
5_cts	36574.0	50%	6.47	0.02
6_global_route	36606.0	50%	6.41	0.02
7_final	36606.0	50%	6.31	0.02

Worst reported setup path: dp (input port clocked by core_clk) to _33164_ (rising edge-triggered flip-flop clocked by core_clk). Clock source/target latencies and the skew term are retained in the saved stage timing reports and final flow log. Setup uncertainty remains 0.2 ns.

Positive slack means the checked paths satisfy the specified single-corner constraints. The hold result applies to a 0.05 ns hold uncertainty; the original blanket 0.2 ns assumption failed repair. Input/output delays remain 1 ns. Setup uncertainty remains 0.2 ns.

Synthesis has no physical parasitics; placement uses placement estimates; global routing uses guide-based estimates; final timing uses extracted SPEF and propagated clocks. No fast/slow PVT analysis, process-variation derating, crosstalk signoff or package timing is claimed.

Post-route critical timing paths

OpenROAD Timing Report after loading final ODB, propagated-clock constraints and extracted SPEF. The detailed route does not by itself establish multi-corner signoff.

Timing Report

Settings | Columns | Update

Setup | Hold

Capture Clock	Required	Arrival	Slack	Skew	Logic Delay	Logic Depth	Fanout	Start	End
core_clk	10.029	3.716	6.312	-0.065	2.624	62	182	dp	_33164_/D
core_clk	10.029	3.692	6.336	-0.065	2.603	62	180	derr	_33164_/D
core_clk	10.029	3.691	6.337	-0.065	2.602	62	180	derr	_33164_/D
core_clk	10.029	3.691	6.338	-0.065	2.601	62	180	derr	_33164_/D
core_clk	10.029	3.672	6.357	-0.065	2.602	62	184	dr	_33164_/D
core_clk	10.029	3.671	6.358	-0.065	2.578	62	183	dp	_33164_/D
core_clk	10.030	3.669	6.361	-0.066	2.576	60	179	dp	_33163_/D
core_clk	10.029	3.668	6.361	-0.065	2.561	60	182	dp	_33164_/D
core_clk	10.029	3.666	6.363	-0.065	2.573	59	179	dp	_33294_/D
core_clk	10.028	3.664	6.364	-0.065	2.571	60	180	dp	_33196_/D
core_clk	10.024	3.657	6.367	-0.061	2.564	60	179	dp	_33162_/D
core_clk	10.030	3.645	6.385	-0.066	2.555	60	177	derr	_33163_/D
core_clk	10.029	3.642	6.387	-0.065	2.552	59	177	derr	_33294_/D
core_clk	10.028	3.640	6.388	-0.065	2.550	60	178	derr	_33196_/D

Data Path Details | Capture Path Details

Pin	Fanout	⇅	Time	Delay	Slew	Load
dp	1	↓	1.000	1.000	0.100	1.797
clock network delay			1.000	0.000		
input34/A (BUF_X1)		↓	1.000	0.000	0.100	
input34/Z (BUF_X1)	3	↓	1.093	0.093	0.035	36.233
_17290_/A (OAI211_X1)		↓	1.114	0.022	0.046	
_17290_/ZN (OAI211_X1)	1	↑	1.151	0.037	0.023	1.553
_17291_/A2 (AND2_X1)		↑	1.151	0.000	0.023	
_17291_/ZN (AND2_X1)	3	↑	1.211	0.060	0.029	11.546
_17773_/A3 (NAND3_X2)		↑	1.212	0.001	0.029	
_17773_/ZN (NAND3_X2)	2	↓	1.245	0.033	0.021	8.931
_17774_/A (INV_X1)		↓	1.246	0.000	0.021	
_17774_/ZN (INV_X1)	1	↑	1.278	0.032	0.019	7.203
_17793_/A (OAI21_X4)		↑	1.278	0.000	0.019	
_17793_/ZN (OAI21_X4)	3	↓	1.303	0.025	0.016	13.662

Final electrical constraint check

The final extracted-RC checkpoint records the capacitance result after buffering, sizing, routing and extraction. Electrical limits are checked separately from setup/hold and geometry DRC.

```
+ cat build/evidence/electrical_review.txt
POST-ROUTE ELECTRICAL CONSTRAINT CHECK
Source: build/asic/flow.log (final extracted-RC checkpoint)
Units: capacitance in fF; slack is limit minus actual

Maximum-capacitance violations: 0

Design area 36606 u^2 50% utilization.
worst slack 6.31
worst slack 0.02
tns 0.00
Clock core_clk
   0.27 source latency _33452_/CK ^
  -0.25 target latency _33054_/CK ^
   0.20 clock uncertainty
   0.00 CRPR
--------------
   0.23 setup skew

[INFO PSM-0040] All shapes on net VDD are connected.
[INFO PSM-0040] All shapes on net VSS are connected.
FLOW_COMPLETE
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Physical verification and GDS export

KLayout imports routed DEF using the platform layer map, merges the Nangate standard-cell GDS masters and writes rv32_pipeline.gds. Every instantiated master is checked for non-empty geometry. This confirms stream-out completeness, rather than transistor-level equivalence.

Check	Outcome and interpretation
Detailed-router DRC	0 final markers. This uses the router technology model and is distinct from GDS DRC.
Independent KLayout DRC	0 markers using the public FreePDK45 rule deck, including FEOL, BEOL, grid and antenna rule groups.
Power connectivity	VDD/VSS grid checks pass; all 94,104 instance power/ground terminals connect to the expected supply. No supply bump/pad model or voltage-drop analysis was performed.
LVS	PASS: routed processor topology and all 443 named top ports match. An externally pinned FreePDK45 deck was adapted and validated against the installed Nangate cells.
Logical supplemental check	Directed scoreboard passes on mapped/post-route netlists at zero delay. This is not physical LVS or an SDF simulation.

Physical check results

Recorded stream-out and independent DRC status. Any nonzero count remains a reported limitation; gate simulation and power continuity do not replace LVS.

```
+ cat build/evidence/physical_checks_review.txt
PHYSICAL VERIFICATION - independent GDS checks
Command: python3 asic/open/physical_checks.py
Source: reports/physical_checks.json and build/asic/klayout_drc.lyrdb

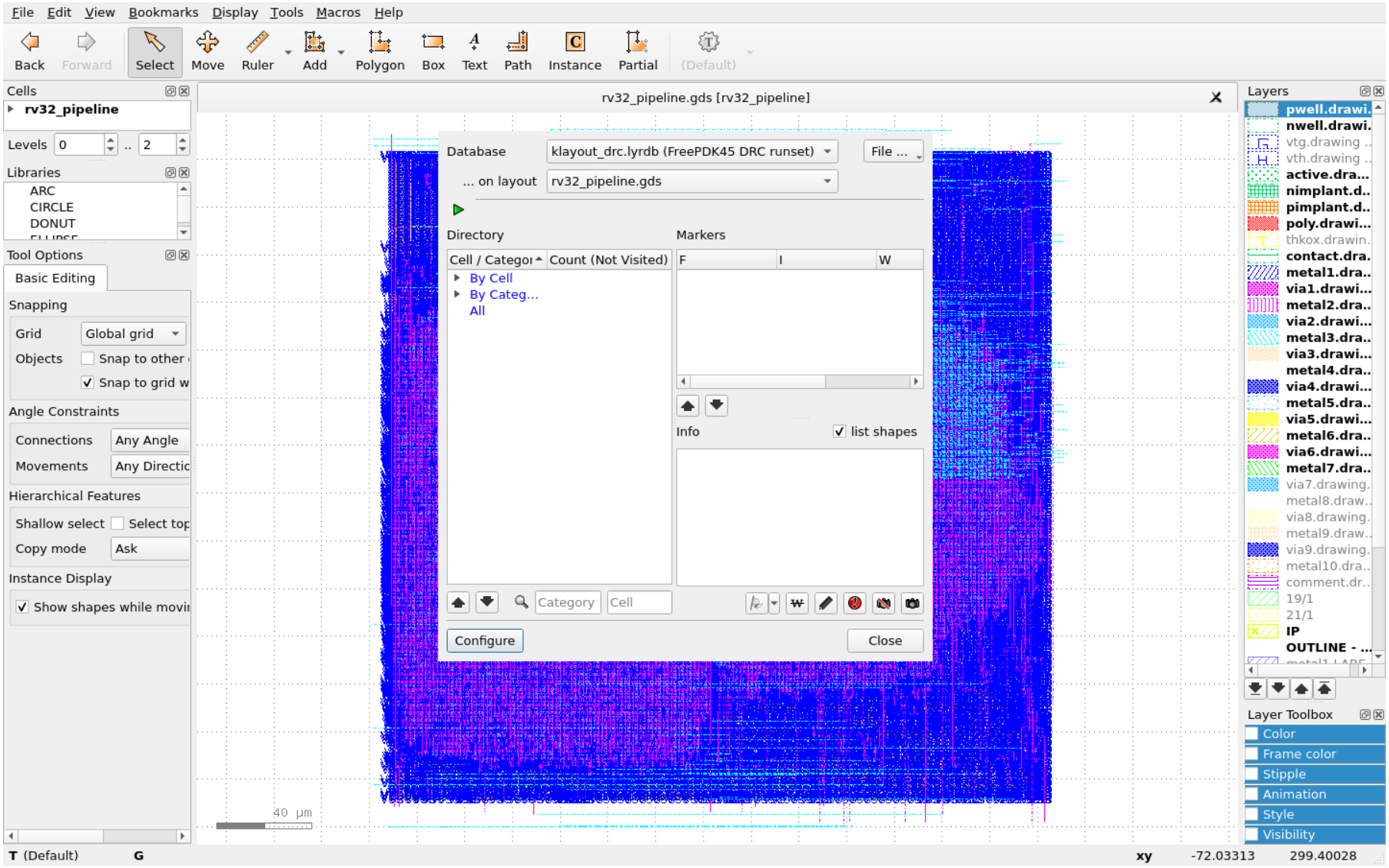
Warning: DEF UNITS does not match reader DBU (DEF UNITS is 2000 and corresponds to a DBU of 0.0005, but reader
set to 0.0001) (line=5, cell=, file=7_final.def)
PASS DEF to GDS stream-out; every cell master has geometry
TOP rv32_pipeline DBU 0.0001 BBOX (0,0;292.425,292.425) CELLS 103

{
  "gds_sha256": "32556ecb66881c322d7e53f3ff99e6e60b24e449638015a4b9adc961f6f54559",
  "drc_deck_sha256": "4e35cad6786f21c6e233fb606bea59bb44e1a12713008cf7367d65a1a3de2147",
  "drc_exit": 0,
  "drc_report": "build/asic/klayout_drc.lyrdb",
  "lvs": "PASS: routed processor topology and all 443 named top ports match",
  "lvs_deck_exists": true,
  "drc_markers": 0,
  "categories": {},
  "lvs_deck": "asic/open/lvs/processor.lvs",
  "lvs_summary": "reports/lvs_summary.json"
}
+ set +x

Recorded log review complete. Original logs retained in build/.
```

GDS and independent DRC in KLayout

Final merged GDS loaded in KLayout, with the independent DRC results database. Marker categories identify the rules requiring review.



Instance power and ground connectivity

Explicit OpenDB audit of every POWER/GROUND instance terminal. Sizing protection is released before final global power connection; fixed placements are preserved. This supplements the PDN shape-continuity check.

```
+ cat build/evidence/power_pins_review.txt
POWER / GROUND INSTANCE TERMINAL AUDIT
Command: openroad -python -exit asic/open/check_power_pins.py
Source: reports/power_pins.json and build/asic/7_final.odb

{
  "status": "PASS",
  "checked_power_ground_terminals": 94104,
  "errors": [],
  "odb_sha256": "14542ddc1e9b80c79bc60beabe0ba13cc6a023cf56c6fc0b6e1419d4d0e8184d"
}

Every POWER terminal must connect to VDD.
Every GROUND terminal must connect to VSS.
Buffer sizing protection is released before final global_connect.
PDN shape continuity and transistor-level LVS are checked separately.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

LVS flow and validation

KLayout independently extracts transistor devices and routed connectivity from the final GDS. OpenROAD write_cdl exports the reference connectivity from the final database using the Nangate library pin order; the independent transistor-level library CDL is explicitly included.

The public FreePDK45_for_KLayout deck is pinned at d15da945559773854ca04aeb9bc2c150d55a1e6c, with the original BSD-3-Clause source and license retained. The adaptation removes unrelated OpenRAM cheat blocks, corrects MOS tolerance class names and adds an explicit verdict and named-port check.

Validation	Measured outcome
Basic positive fixtures	INV_X1 and DFF_X1: PASS
Transistor-width negative control	Double the inverter MOS widths: expected FAIL, observed FAIL
Port-label negative control	Swap A and ZN labels without altering geometry: expected FAIL, observed FAIL
Instantiated cell-master sweep	83 functional masters: all PASS
Comparison validity	Nonempty processor and cell circuits; all circuit pairs and top-level relation pairs match.

MOS W/L tolerance: 1 nm absolute plus 0.1%. The deck follows the Nangate implicit-well convention: PWELL/BULK bind to VSS and NWELL to VDD. Only those well aliases are canonicalized; signal labels retain identity, with case normalized for SPICE. No processor or functional-cell black boxes are used.

LVS deck validation and negative controls

Positive checks must pass and intentionally incorrect references/layout labels must fail. The whole set of instantiated standard-cell masters is independently checked.

```
+ cat build/evidence/lvs_cells_review.txt
LVS DECK VALIDATION - real KLayout comparisons
Source: reports/lvs_cells.json and lvs_library.json
Command: python3 asic/open/lvs/run.py
Library sweep: python3 asic/open/lvs/validate_library.py

INV_X1: PASS (expected PASS)
DFF_X1: PASS (expected PASS)
INV_X1_negative: FAIL (expected FAIL)
INV_X1_labels_negative: FAIL (expected FAIL)

All 83 instantiated functional masters: PASS

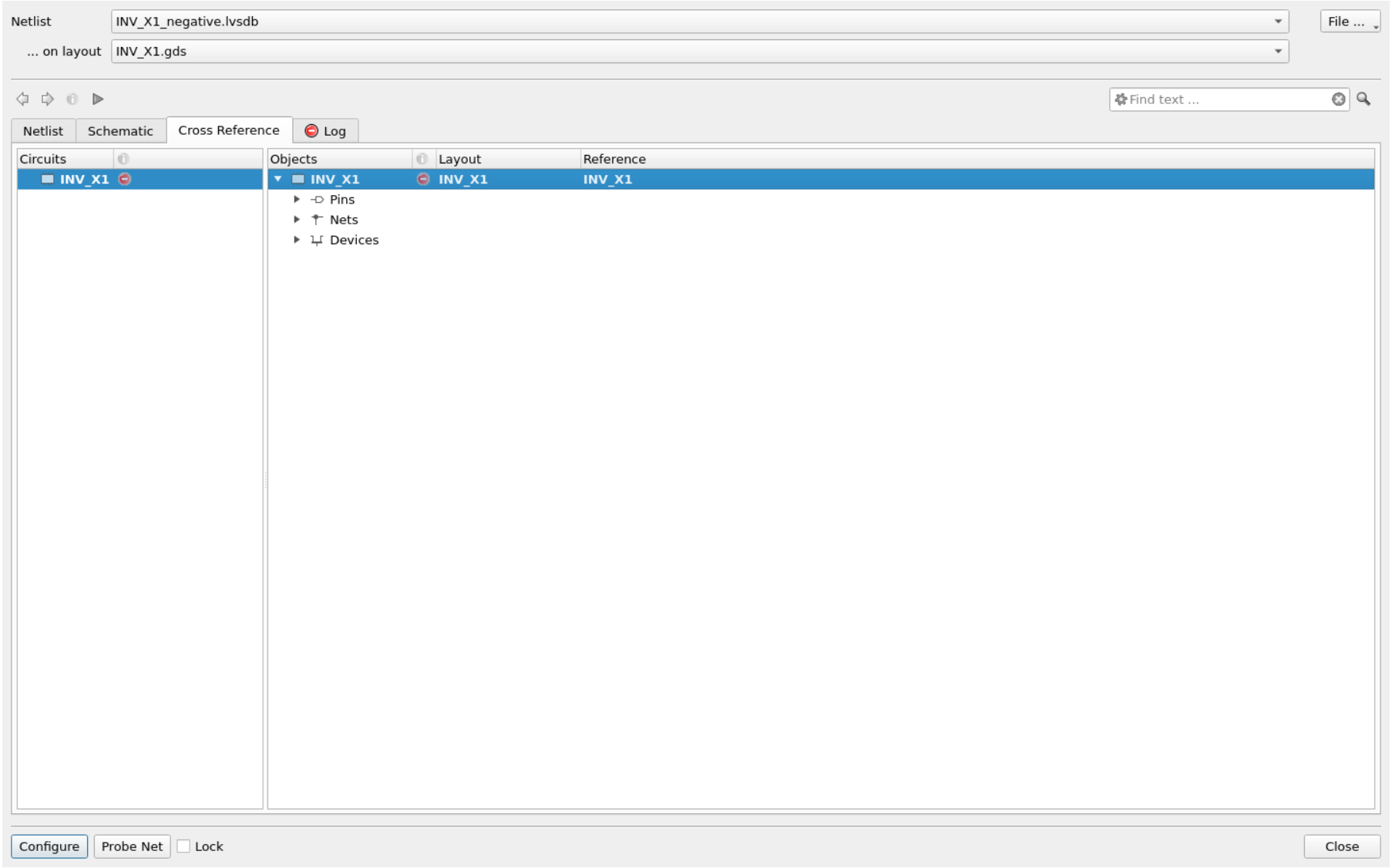
Controls deliberately change transistor width and swap signal labels.
Both must be rejected; a zero tool exit code alone is not a pass.

Deck: pinned FreePDK45_for_KLayout, BSD-3-Clause.
Adaptations: PWELL to VSS, well-alias names, SPICE case normalization,
correct MOS tolerance class names, explicit compare and port checks.
No functional black boxes or processor cheat blocks.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Negative control rejected by LVS

The actual KLayout cross-reference browser shows a mismatch for the inverter whose reference transistor widths were deliberately doubled. This verifies that the flow detects a known defect.



Processor LVS results

Metric	Result
Overall comparison	PASS
Matched circuit pairs	84 (functional cell masters plus processor)
Matched named top-level pins	443
Matched top-level nets	21593
Matched standard-cell instances	19030
Top-level port identities	PASS after SPICE letter-case normalization; bus indices preserved
Comparison duration	357.65 seconds
Comparison log	0 errors; 108 internal-net ambiguity warnings
Cross-reference database	build/lvs/processor.lvsdb

All compared circuits and top-level relation pairs report Match. The warning count above concerns ambiguous internal net assignments; every circuit still reports Match and no comparison errors remain. LVS does not verify path delay, antenna limits or maximum-capacitance limits.

Routed processor LVS and named-port audit

Actual recorded verdict and audited comparison counts. Input fingerprints in lvs_summary.json bind this result to the checked GDS, reference netlist and final OpenROAD database.

```
+ cat build/evidence/lvs_processor_review.txt
ROUTED PROCESSOR LVS - PASS
Command: python3 asic/open/lvs/run.py --processor
Audit: klayout -b -r asic/open/lvs/audit.py
Source: reports/lvs_summary.json; build/lvs/processor.lvsdb

Matched circuit pairs: 84
Matched top pins: {'Match': 443}
Matched top nets: {'Match': 21593}
Matched cell instances: {'Match': 19030}
Named port-label check: True
Cell masters independently validated: 83
Comparison duration: 357.65 seconds

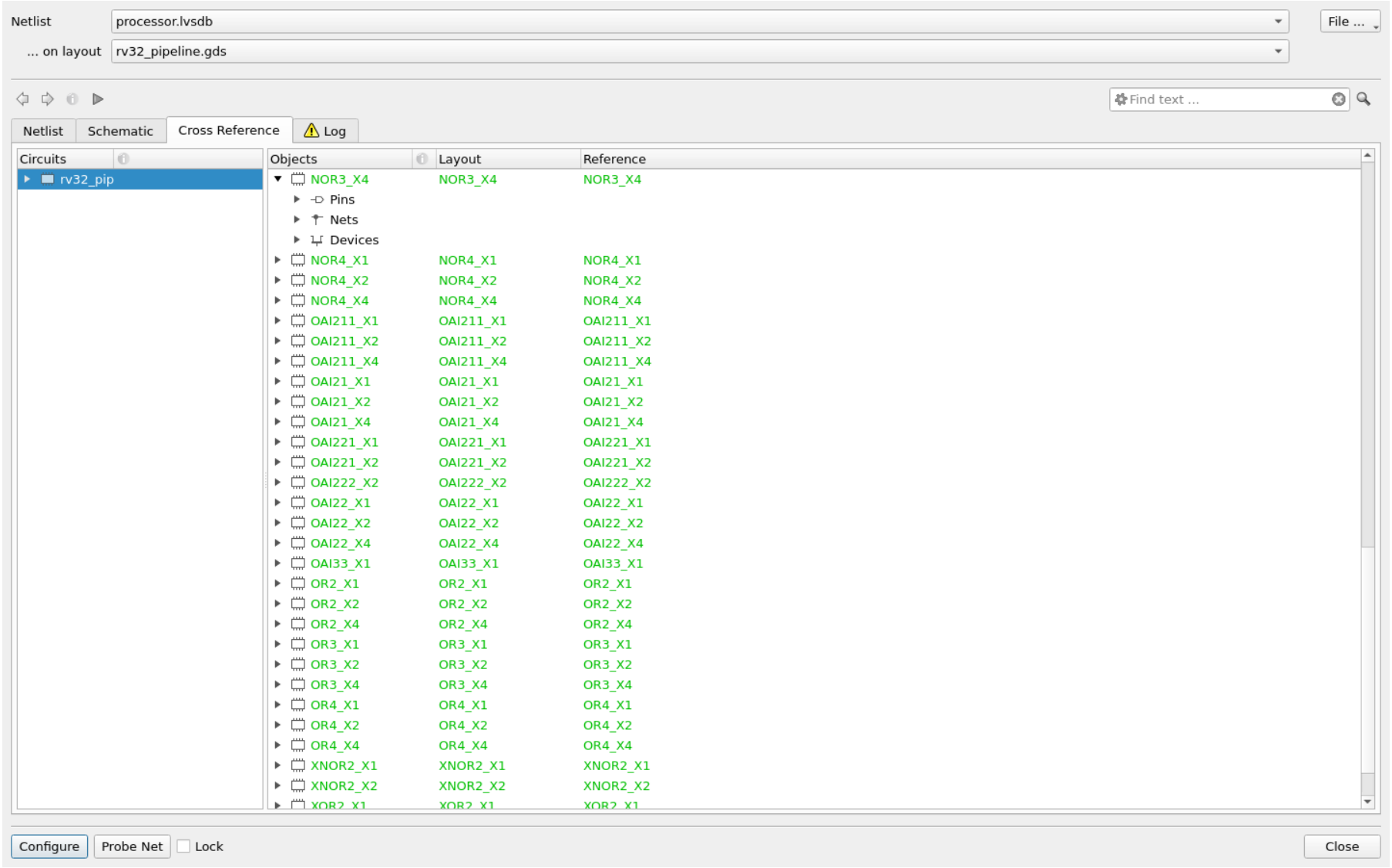
LVS_RESULT PASS

Geometry DRC / electrical checks remain separate:
0 independent DRC markers and 0 maximum-capacitance violations; academic PDK flow, not foundry signoff
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Processor LVS cross-reference in KLayout

Actual native LVS database browser. Green circuit relations indicate matched extracted and reference circuits. The saved database supports further net, pin, device and instance inspection.



Antenna and electrical closure

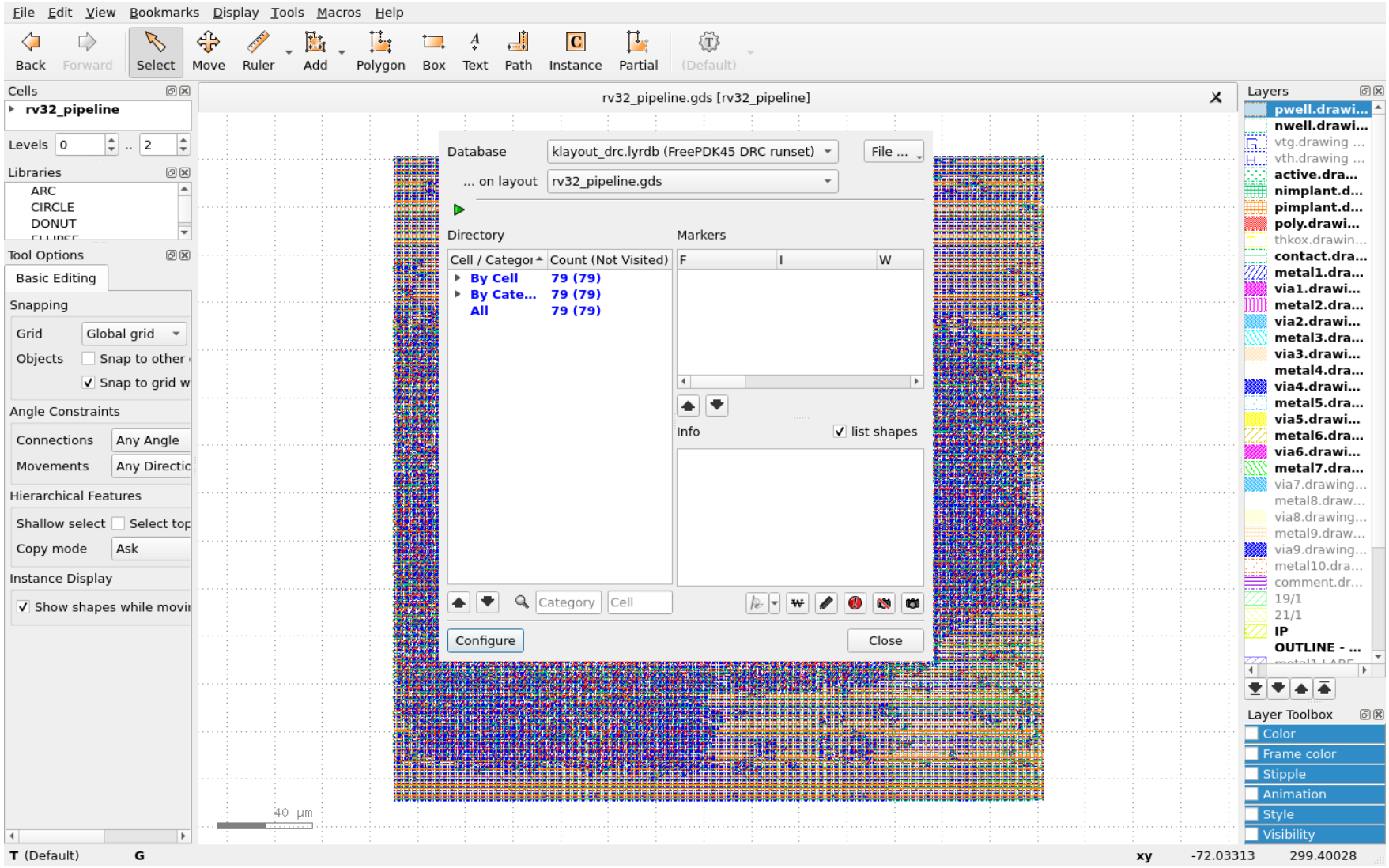
Check	Original routed core	Repaired routed core
Independent GDS antenna markers	43	0
Maximum-capacitance violations	8; worst excess 11.10 fF	0
Cell area (um2; tool-rounded)	36,027	36606.0
Detailed-router markers	0	0
Extracted setup / hold slack	+6.44 / +0.02 ns	+6.31 / +0.02 ns
LVS	PASS before repair	PASS on the new GDS

Repair anchors real input buffers at legal sites near their core pins, protects them against removal/resizing, and anchors the CTS root near clk. It restricts signal/clock routing to metals 2-6/4-6, and applies electrical repair using global-route parasitics with a 20% capacitance margin and 10% slew margin. Cells are legalized and affected nets re-routed before detailed routing. Final extraction, GDS DRC, routed netlist simulation and transistor-level LVS are repeated.

The original Nangate antenna cell has zero declared diffusion area and an empty CDL device body. It was not used as an unverified diode repair. No DRC rules or Liberty capacitance limits were relaxed, and no antenna markers were waived.

Intermediate repair: antenna check still fails

The first candidate cleared capacitance but left small-input-gate buffers free to move toward internal loads. The independent deck reported 79 antenna markers, so this candidate was rejected. The final repair anchors and protects the input buffers and CTS root.



Rejected candidate - measured check results

Actual saved DRC and electrical results for the intermediate candidate. These are repair-history measurements and are not the final submitted layout result.

```
+ cat build/evidence/antenna_candidate_review.txt
REJECTED FIRST REPAIR CANDIDATE - ACTUAL CHECK RESULTS
Command: python3 asic/open/physical_checks.py
Sources retained: build/closure_candidate_1/

Independent GDS DRC markers: 79
{
  "METAL3_ANTENNA": 18,
  "METAL4_ANTENNA": 26,
  "METAL5_ANTENNA": 19,
  "METAL6_ANTENNA": 16
}

Final capacitance violations: 0
Final setup/hold slack: 6.4 / 0.02 ns

This candidate was rejected because antenna markers remained.
The input buffers were subsequently anchored and protected.
The final layout is checked separately.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Electrical repair commands and tool output

Recorded OpenROAD repair output and the exact buffering, routing and margin commands from the reproducible physical flow.

```
+ cat build/evidence/repair_review.txt
ANTENNA / ELECTRICAL REPAIR - actual flow log excerpts
Command: python3 asic/open/run_physical.py
Source: asic/open/flow.tcl and build/asic/flow.log

buffer_ports -inputs -buffer_cell BUF_X8
set_routing_layers -signal metal2-metal6 -clock metal4-metal6
repair_design -cap_margin 20 -slew_margin 10
pin_input_buffers (legal edge sites; protected against resizing)
(placement and global-route estimates; legalized before route)

[INFO RSZ-0042] Inserted 32 tie LOGIC0_X1 instances.
[INFO RSZ-0027] Inserted 73 input buffers.
[INFO RSZ-0036] Found 93 capacitance violations.
[INFO RSZ-0038] Inserted 288 buffers in 4060 nets.
[INFO RSZ-0039] Resized 3213 instances.
[INFO RSZ-0036] Found 24 capacitance violations.
[INFO RSZ-0038] Inserted 5 buffers in 12 nets.
[INFO RSZ-0039] Resized 1 instances.

Final capacitance violations: 0
Final setup/hold: 6.31 / 0.02 ns
Final independent GDS DRC and LVS are separately audited.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Final physical closure audit

The audit requires current GDS/ODB/deck fingerprints, zero GDS and router markers, zero capacitance violations, nonnegative setup/hold slack, passing routed simulation, power connectivity and passing LVS.

```
+ cat build/evidence/closure_review.txt
PHYSICAL CLOSURE - CHECKED FINAL ARTIFACTS
Command: python3 asic/open/closure.py
Source: reports/closure_summary.json

Original: 43 antenna markers; 8 capacitance violations.

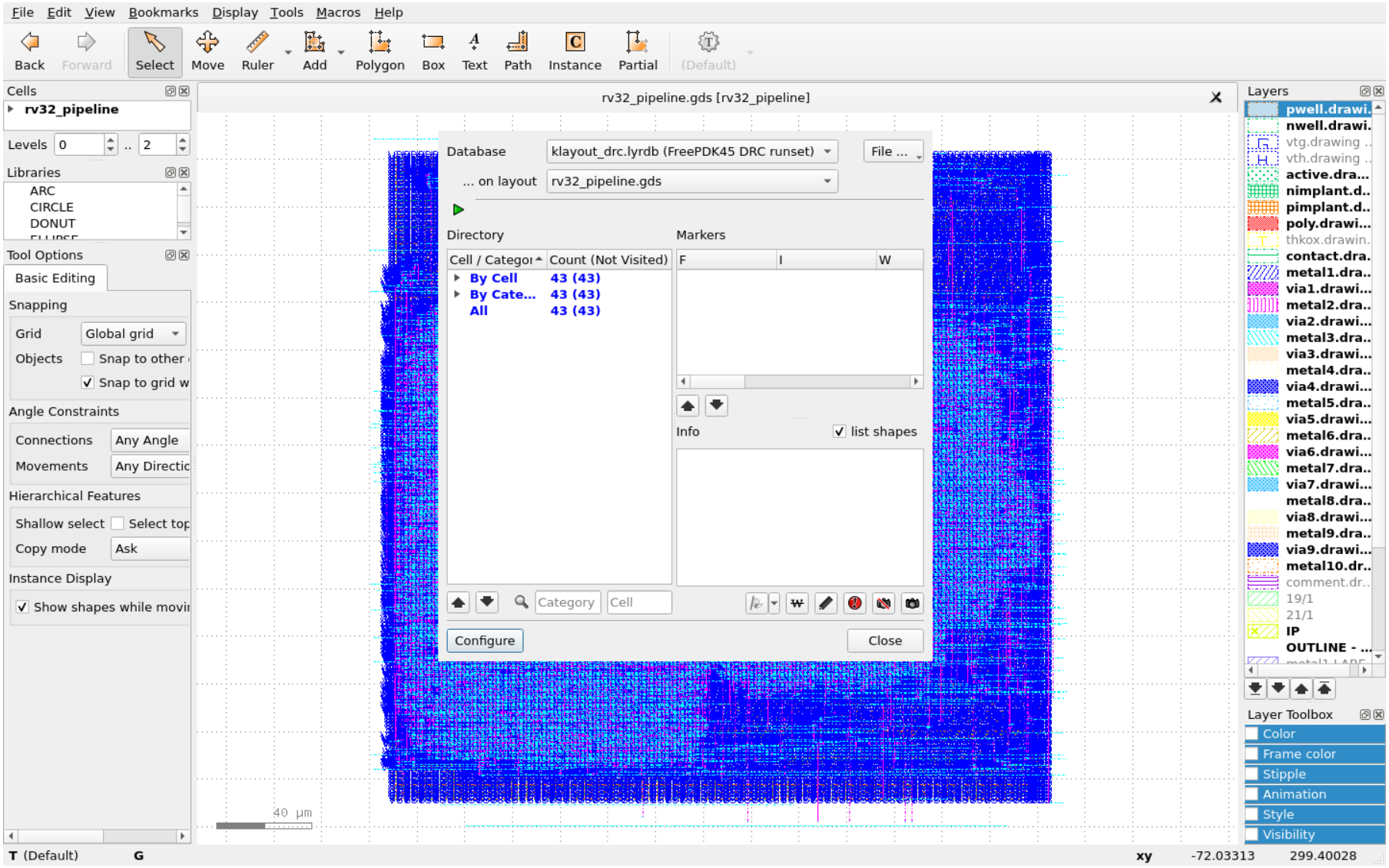
FINAL CHECKS
{
  "anchored_primary_inputs": 74,
  "max_anchor_distance_um": 20.88,
  "gds_drc_markers": 0,
  "cell_area_um2_rounded": 36606.0,
  "router_drvs": 0,
  "maximum_capacitance_violations": 0,
  "setup_wns_ns": 6.31,
  "hold_wns_ns": 0.02,
  "lvs": "PASS",
  "gate_scoreboard": "PASS",
  "power_connectivity": "VDD and VSS PASS"
}

No DRC rules or Liberty capacitance limits were relaxed.
Full GDS LVS repeated after the physical changes.
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Original antenna markers before repair

Preserved KLayout view of the original 43-marker result. The final repeated independent check appears earlier in this report.



Original capacitance violations before repair

Preserved original extracted result: eight violating pins, with worst excess 11.10 fF. The final electrical check appears earlier in this report.

```
+ cat build/evidence/electrical_review.txt
POST-ROUTE ELECTRICAL CONSTRAINT VIOLATIONS
Source: build/asic/flow.log (last max capacitance report)
Units: capacitance in fF; slack is limit minus actual



Pin                                   Limit     Cap   Slack
-----------------------------------------------------------
_21957_/Z                            120.85  131.95  -11.10 (VIOLATED)
_18746_/ZN                            41.50   47.14   -5.64 (VIOLATED)
_25003_/ZN                           233.15  237.43   -4.28 (VIOLATED)
_17840_/ZN                           241.70  245.66   -3.96 (VIOLATED)
_17298_/ZN                           106.81  108.07   -1.26 (VIOLATED)
_33060_/Q                            121.15  121.79   -0.64 (VIOLATED)
_33056_/Q                            121.15  121.27   -0.11 (VIOLATED)
_20569_/ZN                           106.81  106.83   -0.02 (VIOLATED)

[INFO PSM-0040] All shapes on net VDD are connected.
[INFO PSM-0040] All shapes on net VSS are connected.
FLOW_COMPLETE
+ set +x

Recorded log review complete. Original logs retained in build/.
```

Implementation scope and remaining limits

The repaired result is assessed against the same enabled public FreePDK45 DRC rules, Nangate Liberty limits and 10 ns clock constraint. Exact check counts, timing and input fingerprints are retained in the attached machine-readable reports.

Hierarchical transistor-level LVS passes under the documented public-PDK extraction and well-bias assumptions. External memory, a pad ring, package power entry and a physical SRAM implementation remain outside this core project.

Timing and area are educational typical-corner measurements. The disclosed hold uncertainty is 0.05 ns. Multi-corner timing, voltage-drop analysis and power consumption with representative activity have not been performed. The exported GDS is an academic routed core implementation; this result does not establish foundry signoff.

Contributions, scope and reproduction

Contributor / assistance	Recorded role
Shiv Sriram V, 24BVD0168	Project owner; supplied the processor proposal and existing project repository. No additional team members were provided.
Existing repository	Processor, optional SoC extensions, firmware and regression infrastructure predated this session.
AI-assisted work in this session	Open-source floorplan, PDN, placement, CTS, routing, extraction, GDS export, antenna/electrical repair, independent DRC/LVS, power audit and evidence/report preparation; tool execution, evidence capture and report compilation. Student review and approval of the submitted claims remain with the project owner.

No commits were made or staged. Suggested stage-by-stage commit messages and file groupings are in docs/COMMIT_GUIDE.md. The project includes exact constraints, source scripts, tool outputs and screenshots for independent reproduction.

```
make pipeline
make all
python3 asic/open/synth.py
make closure
make evidence
make submissions
```

Source and evidence index

Assignment sources: the supplied Assignment VIII project proposal; ASIC Design Lab QP 9 (1).pdf (Assignment IX, Front End); ASIC Design Lab QP 0.pdf (Assignment X, Back End). Submission 8 duplicates the supplied proposal.

OpenROAD flow and public platform documentation:
https://openroad-flow-scripts.readthedocs.io/en/latest/
Official binary release:
https://github.com/Precision-Innovations/OpenROAD/releases/tag/2024-12-14
LVS source: FreePDK45_for_KLayout - pinned BSD-3-Clause source and license

Evidence location	Content
rtl/ and tb/	Synthesizable sources and independent directed scoreboard
build/evidence/	Full regression, hazard/gate logs and waveforms; terminal review text
build/asic/	Synthesis log/netlist/JSON; ODB, DEF, SDC, STA paths, SPEF, GDS and DRC database
reports/asic_metrics.json	Measured area, utilization, setup/hold slack and tool/platform versions
reports/physical_checks.json	Independent GDS DRC and measured LVS status
reports/lvs_summary.json	Validated cell masters, negative controls and processor comparison counts
reports/closure_summary.json	Final DRC, electrical, timing, functional, power and LVS audit
reports/evidence/	Unaltered actual GUI and xterm screenshot PNG files
asic/open/README.md	Reproduction, constraints, scope and tool setup